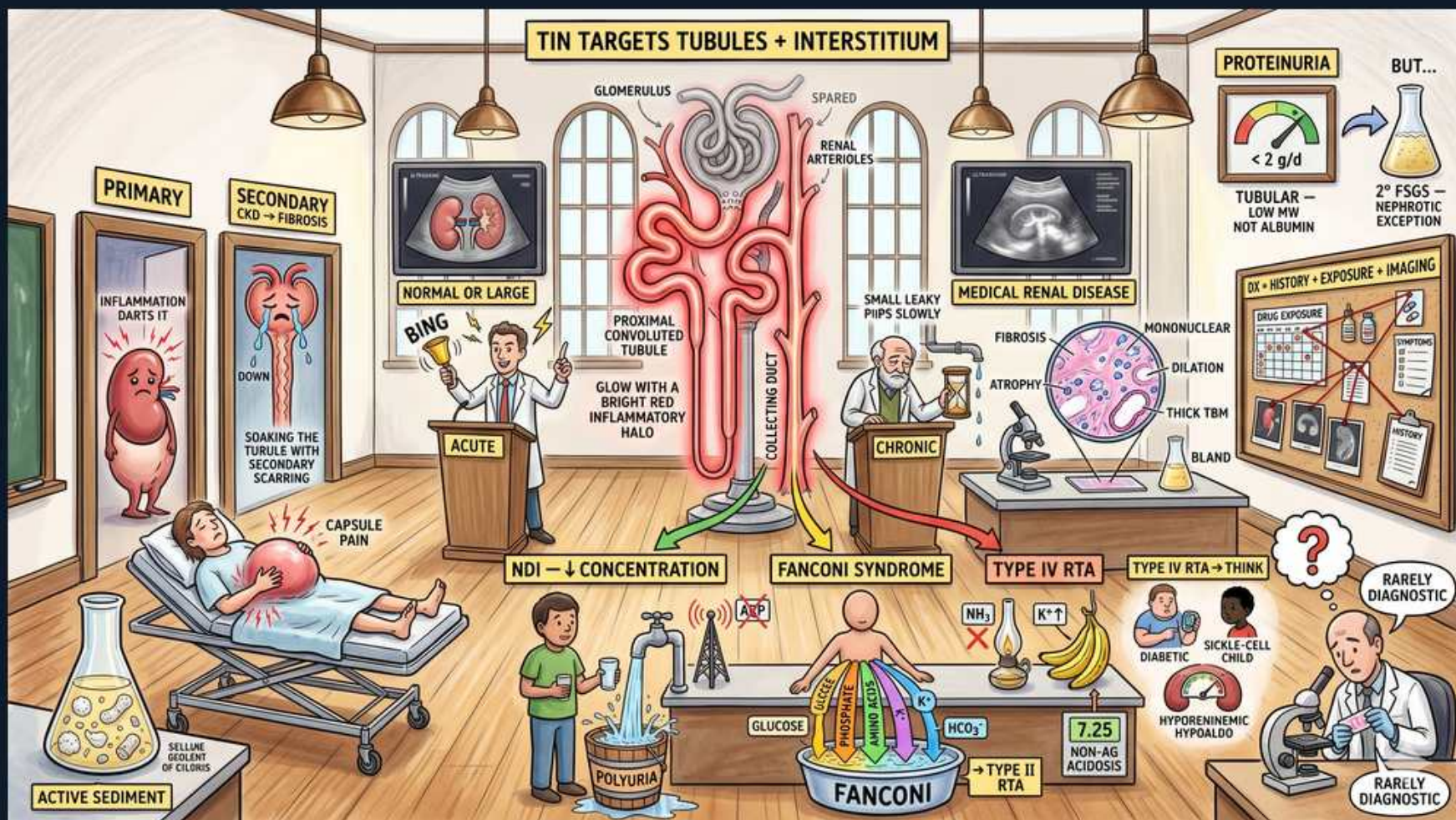


# Tubulointerstitial — TID overview



## 1. TIN targets tubules + interstitium

### WHAT YOU SEE

The top banner reading TIN TARGETS TUBULES + INTERSTITIUM — it names the whole scene's theme: the disease hits tubules and interstitium while the glomeruli stay innocent bystanders.

### WHAT IT ENCODES

Tubulointerstitial disease = inflammation/scarring of the renal tubules and surrounding interstitium with the glomeruli and vessels relatively spared — the inverse anatomic footprint of glomerulonephritis. Clinically it presents as a non-oliguric or non-nephrotic AKI/CKD with tubular dysfunction (RTA, Fanconi, concentrating defect) rather than the hematuria/heavy proteinuria of GN. Mnemonic: 'TIN = Tubules INflamed/scarred, glomeruli innocent bystanders.'

### BOARD TRAP

A vignette with AKI, sterile pyuria and WBC casts but only modest sub-nephrotic proteinuria (<2 g/d) is steered toward 'rapidly progressive glomerulonephritis' or 'nephritic syndrome.' Wrong — RBC casts and dysmorphic RBCs with nephrotic/nephritic-range protein point to glomerular disease; bland-to-WBC sediment with low-grade tubular proteinuria is interstitial. The discriminator is the cast/cell type and the proteinuria magnitude, not the mere presence of AKI.

## 2. Glomeruli/arterioles SPARED

### WHAT YOU SEE

A SPARED label stamped over the drawn glomerulus and renal arterioles — the 'spared' tag IS the key point: the filter is intact, so no RBC casts or heavy albuminuria.

### WHAT IT ENCODES

Glomeruli and renal arterioles are histologically preserved early in TIN — this is the single most useful contrast with glomerulonephritis and vasculitis. Because the filtration apparatus is intact, you get little albuminuria and no RBC casts; the lesion is downstream in the tubules/interstitium. Late/secondary disease can produce secondary FSGS once enough nephrons scar.

### BOARD TRAP

The board offers 'crescentic glomerulonephritis' or 'pauci-immune vasculitis' for a patient with biopsy showing dense interstitial lymphocyte/eosinophil infiltrate but normal glomeruli. Wrong — normal glomeruli on light microscopy with interstitial inflammation defines AIN; crescents and necrotizing capillary loops would be needed for crescentic GN.

## 3. Acute vs Chronic

### WHAT YOU SEE

An ACUTE chalkboard on the left versus a CHRONIC hourglass on the right — the chalkboard (days-weeks, drug-induced) versus the draining hourglass (months-years of fibrosis) IS the acute-vs-chronic time axis.

### WHAT IT ENCODES

Acute interstitial nephritis (AIN) develops over days–weeks (most commonly drug-induced: classic triad of fever, rash, eosinophilia/eosinophiluria, though the full triad appears in <10%) with edematous interstitium and inflammatory infiltrate. Chronic interstitial nephritis (CIN) evolves over months–years with interstitial fibrosis, tubular atrophy and thickened tubular basement membranes, and is usually irreversible. Time course on the vignette (acute creatinine bump after a new drug vs years of slow decline) is the key sorting tool.

### BOARD TRAP

A patient on a PPI/NSAID for years presenting with slowly rising creatinine is labeled 'acute drug AIN, will fully reverse with steroids.' Wrong — long-standing exposure with fibrosis on biopsy is chronic interstitial nephritis, where steroids give little benefit and recovery is incomplete; the discriminator is chronicity/fibrosis vs an acute cellular infiltrate.

## 4. Primary vs Secondary

### WHAT YOU SEE

Two green boards labelled PRIMARY and SECONDARY — they split disease starting in the tubules from interstitial scarring that is the downstream end-point of every chronic kidney disease.

### WHAT IT ENCODES

Primary tubulointerstitial disease starts in the tubules/interstitium (drugs, autoimmune AIN, toxins, infection). Secondary tubulointerstitial scarring is the downstream final common pathway of nearly every chronic kidney disease — chronic GN, diabetic and hypertensive nephrosclerosis all end in interstitial fibrosis, and the degree of interstitial fibrosis/tubular atrophy (IFTA) on biopsy is the strongest histologic predictor of long-term GFR. So 'how much interstitial scarring' prognosticates better than the glomerular lesion.

### BOARD TRAP

A diabetic with proteinuria and interstitial fibrosis is called 'primary chronic interstitial nephritis.' Wrong — the fibrosis is secondary to diabetic nephropathy; the discriminator is the dominant primary lesion (Kimmelstiel-Wilson nodules, GBM thickening) driving the interstitial change.

## 5. Proximal tubule + inflammatory halo

### WHAT YOU SEE

A glowing proximal convoluted tubule ringed by a bright red inflammatory halo — the red halo IS the T-cell interstitial infiltrate of AIN crowding the hardest-working tubule.

### WHAT IT ENCODES

The proximal convoluted tubule, with the highest metabolic and transport workload, is the prime target for ischemic and toxic injury and concentrates filtered toxins/drugs/light chains. In AIN the interstitium surrounding the tubules fills with T lymphocytes (cell-mediated, type IV hypersensitivity), plus eosinophils and sometimes granulomas; tubulitis = inflammatory cells crossing the tubular basement membrane is the histologic hallmark.

### BOARD TRAP

A vignette shows interstitial eosinophils and the answer key tempts 'IgE-mediated type I hypersensitivity / give antihistamines.' Wrong — drug AIN is predominantly a T-cell-mediated (type IV) delayed hypersensitivity; management is stop the offending drug and consider glucocorticoids, not antihistamines.

## 6. Normal/large kidneys (acute)

### WHAT YOU SEE

Two plump kidneys labelled NORMAL OR LARGE — the full-size kidneys ARE the acute, potentially reversible process (edema/infiltrate), in contrast to small scarred chronic kidneys.

### WHAT IT ENCODES

In acute interstitial nephritis kidneys are normal-sized or enlarged and may be echobright due to interstitial edema and cellular infiltrate; gallium-67 scan can light up. In chronic disease they shrink and scar. Kidney size on ultrasound is therefore a quick acuity/chronicity clue: big/normal = potentially reversible acute process, small/echogenic = chronic, irreversible.

### BOARD TRAP

Small echogenic kidneys are attributed to 'acute interstitial nephritis, treat aggressively for recovery.' Wrong — small kidneys signal chronicity and irreversible scarring; AIN keeps kidneys normal or enlarged. The discriminator is bilateral small echogenic kidneys = chronic, generally do not expect functional recovery.

## 7. Medical renal disease / echogenic

### WHAT YOU SEE

An ultrasound captioned 'small leaky pipes slowly' under a MEDICAL RENAL DISEASE label — the small bright kidney on US IS the chronic medical-renal pattern of fibrosis with bland urine.

### WHAT IT ENCODES

Chronic interstitial disease on ultrasound = bilateral small kidneys with increased cortical echogenicity ('medical renal disease' pattern), reflecting interstitial fibrosis, tubular atrophy and thickened tubular basement membranes. The urine is characteristically bland (few cells/casts) with low-grade tubular proteinuria, distinguishing it from active glomerular sediment. CIN often presents as slowly progressive CKD with disproportionate tubular dysfunction.

### BOARD TRAP

Bilateral small echogenic kidneys with bland urine are worked up as 'obstructive uropathy, order urgent nephrostomy.' Wrong — obstruction causes hydronephrosis/dilated collecting system, not small echogenic kidneys; the echogenic non-dilated pattern is chronic medical renal (interstitial) disease.

## 8. Tubular proteinuria <2 g/d

### WHAT YOU SEE

A proteinuria gauge needle sitting below 2 g/d next to a low Hb/albumin bottle — the low-reading gauge IS sub-nephrotic, low-molecular-weight tubular proteinuria, not heavy albuminuria.

### WHAT IT ENCODES

Tubular proteinuria is low-molecular-weight ( $\beta$ 2-microglobulin, retinol-binding protein, light chains) because damaged tubules fail to reabsorb filtered small proteins; total is usually <1–2 g/day and NOT albumin-predominant. A urine dipstick (detects albumin) can be near-negative while the sulfosalicylic acid test or a protein/creatinine ratio is elevated — a classic clue to a tubular/overflow source.

### BOARD TRAP

Nephrotic-range proteinuria (>3.5 g/d) is offered as 'consistent with interstitial nephritis.' Wrong — heavy albuminuria points to glomerular disease; interstitial proteinuria is sub-nephrotic and LMW-predominant. The one exception is secondary FSGS from nephron loss, which can reach nephrotic range late.

## 9. Active sediment

### WHAT YOU SEE

A flask labelled ACTIVE SEDIMENT holding cellular casts — the casts in the flask ARE the sterile pyuria and WBC casts of acute interstitial nephritis.

### WHAT IT ENCODES

Acute interstitial nephritis classically gives sterile pyuria (WBCs without bacteria), WBC casts, and sometimes eosinophiluria (Hansel/Wright stain; low sensitivity and specificity — has fallen out of favor). The presence of WBC casts localizes inflammation to the kidney parenchyma rather than the bladder. Hematuria is usually mild and non-dysmorphic; gross hematuria/RBC casts should redirect you to glomerular disease.

### BOARD TRAP

Sterile pyuria with WBC casts is diagnosed as 'urinary tract infection, start antibiotics.' Wrong — sterile pyuria (pyuria with negative culture) plus a recent culprit drug points to AIN; antibiotics (themselves a cause of AIN) won't help and may worsen it. The discriminator is the negative culture and the temporal drug link.

## 10. NDI — ↓concentration

### WHAT YOU SEE

A man pouring out water under an NDI / POLYURIA label — the endless water pour IS the failed concentrating mechanism: nephrogenic DI with dilute urine that ignores desmopressin.

### WHAT IT ENCODES

Interstitial/tubular injury blunts the medullary concentrating mechanism (damaged collecting duct, washed-out medullary gradient) producing acquired nephrogenic diabetes insipidus: polyuria, nocturia, low urine osmolality that fails to rise after desmopressin. Lithium and hypercalcemia are classic causes. Impaired concentrating ability is often the earliest detectable tubular defect.

### BOARD TRAP

Polyuria with dilute urine unresponsive to desmopressin is called 'central diabetes insipidus, give DDAVP.' Wrong — failure to concentrate AFTER desmopressin defines nephrogenic DI (tubular/interstitial); central DI would correct with DDAVP. The water-deprivation/desmopressin response is the discriminator.

## 11. Fanconi syndrome

### WHAT YOU SEE

A FANCONI figure spilling glucose, HCO<sub>3</sub>, phosphate and amino acids out of his pockets — the leaking molecules ARE generalised proximal-tubule wasting (glucosuria, phosphaturia, type II RTA).

### WHAT IT ENCODES

Generalized proximal tubule dysfunction = Fanconi syndrome: wasting of glucose (glucosuria with normal serum glucose), phosphate (hypophosphatemia, osteomalacia), bicarbonate (type II/proximal RTA), amino acids and uric acid. Adult causes include multiple myeloma/light chains, tenofovir, ifosfamide, cisplatin, aminoglycosides, and heavy metals (lead, cadmium). The fingerprint clue is glucosuria with a normal blood glucose.

### BOARD TRAP

A patient with glucosuria, hypophosphatemia and non-anion-gap acidosis is labeled 'uncontrolled diabetes.' Wrong — normoglycemic glucosuria plus phosphaturia and proximal (type II) RTA is Fanconi syndrome; in an adult on tenofovir or with monoclonal protein, think drug/light-chain proximal tubulopathy, not diabetes.

## 12. Type IV RTA

### WHAT YOU SEE

A TYPE IV RTA panel with NH<sub>4</sub> down, K<sup>+</sup> up and a pH 7.25 dial — the high-K, low-ammonium, mildly acidic dial IS hyperkalemic distal RTA from aldosterone deficiency/resistance.

### WHAT IT ENCODES

Type IV (hyperkalemic distal) RTA results from aldosterone deficiency or resistance → reduced distal Na<sup>+</sup> reabsorption and impaired H<sup>+</sup>/K<sup>+</sup> secretion → hyperkalemia and a hyperchloremic non-anion-gap metabolic acidosis with a paradoxically acid urine (impaired ammoniagenesis). Classic settings: diabetic hyporeninemic hypoaldosteronism, chronic interstitial disease, drugs (ACEi/ARB, K-sparing diuretics, trimethoprim, calcineurin inhibitors), and adrenal insufficiency. Treat with low-K diet, loop diuretic, fludrocortisone if appropriate.

### BOARD TRAP

Hyperkalemia with non-gap acidosis is attributed to 'distal (type I) RTA.' Wrong — type I distal RTA causes HYPOkalemia with inability to acidify urine (urine pH >5.5) and nephrocalcinosis; type IV causes HYPERkalemia with low urine ammonium. Serum potassium (high in IV, low in I/II) is the fastest discriminator.